

EcoNext — Order Management, Fulfillment & Physical Shipment Operations Guide

End-to-End Operational Manual: Customer Order Placement → Order Confirmation → Warehouse Picking & Packing → Physical Shipment Provisioning → Carrier Dispatch & Linehaul → Live GPS Telemetry Checkpoints → Out for Delivery → Cryptographic Delivery OTP / Email Dispatch → Verified Customer Handover

Target Audience: Fulfillment Hub Operators, Warehouse Staff, Logistics Drivers, Admin Team	Document Version: 2.4.0 (Production Release)
Core Services: Order Operations (Port 8083), Django Backend, Spring API Gateway (Port 8080)	Event Architecture: Kafka Bus + Redis Cache + STOMP Live WebSocket
Security & Verification: 6-Digit Redis Delivery OTP, JWT RBAC, Full Audit Trail	Date of Publication: September 2026

1. Executive Summary & Architecture Overview

The EcoNext Operations & Fulfillment System provides an enterprise-grade, real-time workflow for handling orders after a customer completes checkout. Built on a resilient microservices architecture, the system combines strict transactional database consistency (MySQL) with sub-second event streaming (Apache Kafka) and push-based real-time telemetry (STOMP WebSockets) to synchronize warehouse staff, logistics linehaul drivers, admin managers, and end customers.

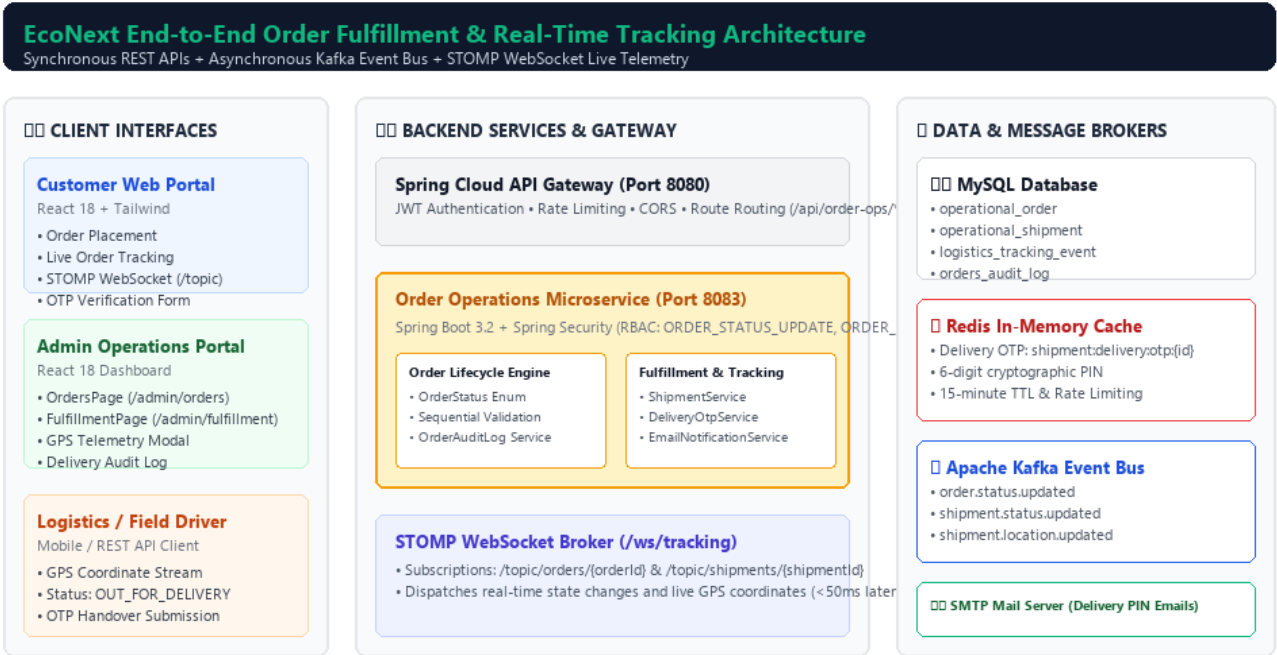


Figure 1.1: High-Level Architecture & Asynchronous Event Streaming Data Pipeline

Key Architectural Pillars:

- **Dual-Entity Separation:** Decouples commercial purchase semantics (OperationalOrder) from physical transit semantics (OperationalShipment). An order can have dedicated shipments and containers.
- **Deterministic State Machines:** Sequential, forward-only status graphs in both backend controllers and frontend portals prevent invalid workflow skips.
- **Cryptographic Delivery Handover:** Zero-trust package delivery verified via 6-digit numeric OTP generated in Redis (15-min TTL) and dispatched through SMTP HTML email.
- **Sub-Second Telemetry Broadcast:** Live vehicle GPS coordinates published via Kafka topic shipment.location.updated and broadcast to STOMP destination /topic/shipments/{id}.
- **Full Audit Compliance:** Every single status change, note, and GPS ping records staff ID, username, and timestamp in MySQL orders_audit_log.

2. End-to-End Operational Lifecycle & State Machines

The EcoNext platform orchestrates order fulfillment through two synchronized state machines: the **Order Lifecycle** and the **Physical Shipment Lifecycle**. Below is the exact transition matrix and operational criteria required for each stage.

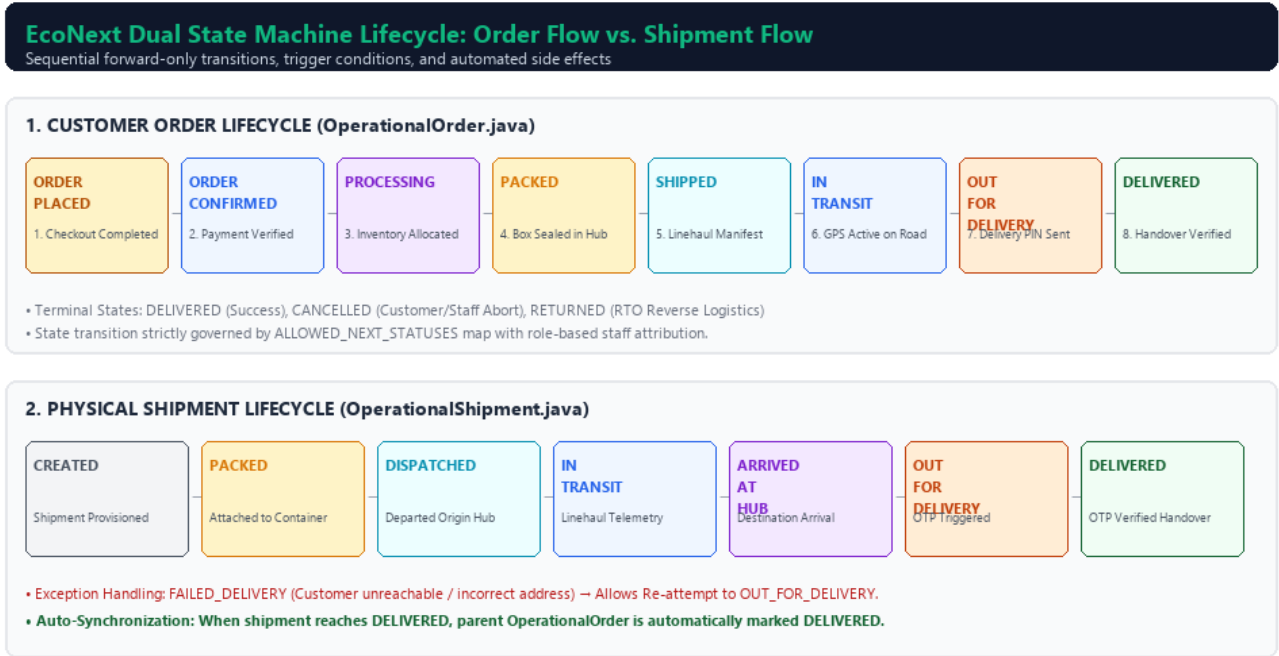


Figure 2.1: Dual State Machine Lifecycle — Order Status vs. Physical Shipment Status

Stage #	Order Status	Shipment Status	Triggering Event / Operational Action	System Side Effects
1	ORDER_PLACED	None	Customer checks out via web portal; initial record saved.	Stock reserved; confirmation notification triggered.

2	ORDER_CONFIRMED	None	Staff verifies payment in OrdersPage.jsx and clicks 'Confirm Order'.	Emits Kafka order.status.updated; moved to warehouse queue.
3	PROCESSING	None	Warehouse operator begins picking inventory items from shelves.	Pick-list generated; barcode validation active.
4	PACKED	CREATED	All SKUs verified in box; Staff creates physical shipment in FulfillmentPage.jsx.	Shipment number #SHP-xxxxx allocated; weight recorded.
5	SHIPPED	DISPATCHED	Package loaded onto linehaul carrier vehicle and departs hub.	Carrier manifest closed; tracking link activated.
6	IN_TRANSIT	IN_TRANSIT / ARRIVED_AT_HUB	Carrier in linehaul transit; GPS telemetry checkpoints submitted.	Live STOMP coordinates pushed to customer map.
7	OUT_FOR_DELIVERY	OUT_FOR_DELIVERY	Shipment assigned to local delivery driver; operator clicks 'Send Out for Delivery'.	DeliveryOtpService generates 6-digit PIN in Redis; sends HTML email.
8	DELIVERED	DELIVERED	Driver/Customer submits valid 6-digit PIN at door.	Redis key cleared; order marked completed; audit log sealed.

3. Role-Based Access Control & Staff Permissions

To ensure compliance, separation of duties, and security, the Order Operations Microservice enforces strict Spring Security method-level authorization (@PreAuthorize).

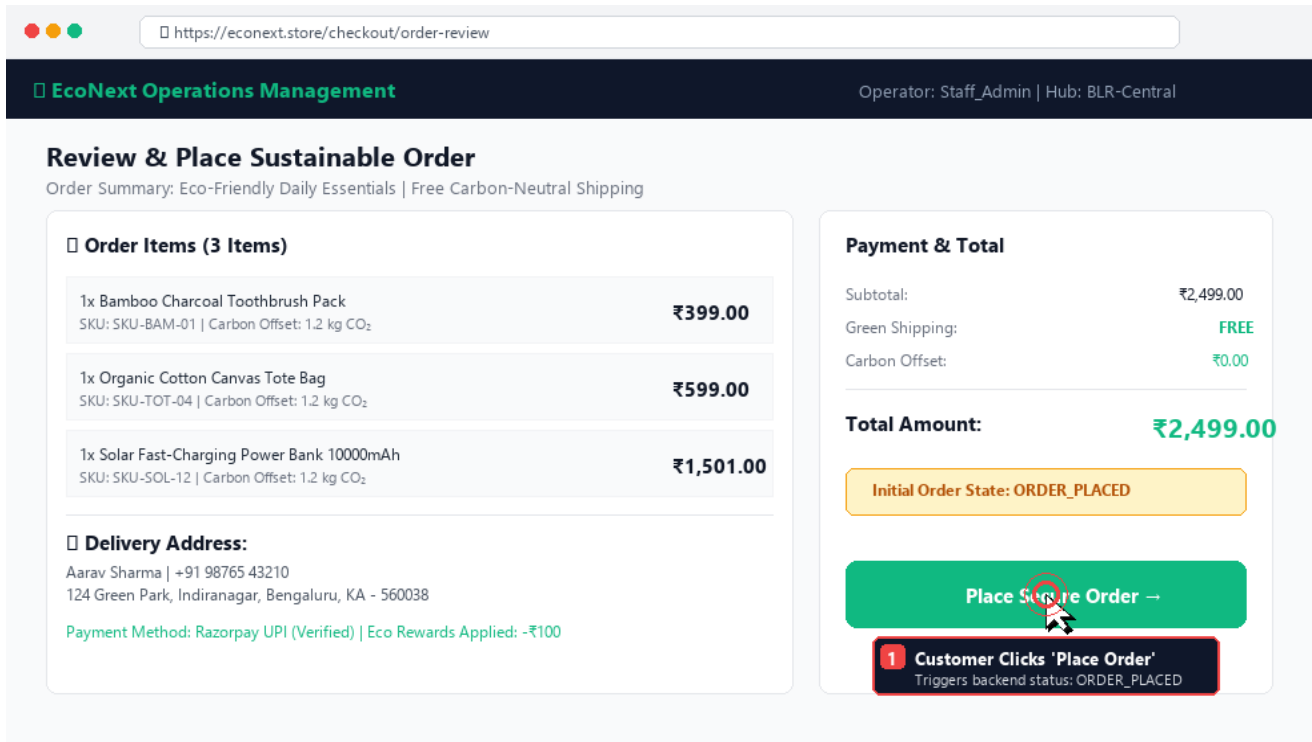
Staff Role	Spring Security Authority	Allowed Portal Functions & Endpoints	Access Restrictions
System Admin	ROLE_ADMIN	Full access to all order status transitions, shipment creation, container management, and manual OTP overrides.	None (Superuser privilege).
Order Manager	ORDER_STATUS_UPDATE , ORDER_READ	Confirm orders, update order status (PATCH /api/order-ops/orders/{id}/status), view customer orders and audit logs.	Cannot delete shipments or alter system configurations.
Warehouse Packing Staff	ORDER_PROCESS, ORDER_READ	Scan inventory items, transition orders to PACKED, create physical shipments (POST /api/order-ops/shipments).	Cannot cancel orders or issue financial refunds.
Logistics / Carrier Agent	ORDER_STATUS_UPDATE	Update GPS coordinates (PATCH /shipments/{id}/location), set OUT_FOR_DELIVERY, submit delivery OTP.	Cannot modify order items, pricing, or customer profiles.

4. Step-by-Step Operator Manual & Web Interfaces

This section details the precise clicks, screen workflows, inputs, and system behaviors for each stage of order fulfillment. Each step features high-resolution web page mockups with red highlighted mouse pointer cursors (■ / ■) and numbered action badges.

Step 1: Customer Order Placement & Checkout (ORDER_PLACED)

Workflow Description: The customer selects sustainable products on the EcoNext store, proceeds to checkout, and completes payment. Upon successful payment authorization, the system assigns a unique reference number (e.g., ORD-10042) and sets the initial state to ORDER_PLACED.



Screenshot 1: Customer Checkout Screen — Mouse Pointer clicking 'Place Secure Order' to initiate ORDER_PLACED state

Operator Verification Checklist for Step 1:

1. Check that the order reference number (ORD-10042) is generated and total matches cart line items.
2. Verify that customer shipping address and phone number are complete.
3. Confirm that payment status is verified via Razorpay webhook or card gateway.

Step 2: Admin Order Confirmation in OrdersPage (/admin/orders)

Workflow Description: Order management staff log in to the EcoNext Admin Portal and navigate to **Orders** (/admin/orders). The operator reviews the incoming orders queue, inspects order items, and clicks the blue **'Confirm Order'** action button. This executes PATCH /api/order-ops/orders/{id}/status with status: ORDER_CONFIRMED.

EcoNext Operations Management Operator: Staff_Admin | Hub: BLR-Central

Customer Order Management

Manage order lifecycles, verify payment confirmation and dispatch to warehouse

Search Order ID, Customer, Phone... **All Orders (48)** **ORDER_PLACED (6)**

ORDER REF	CUSTOMER	ITEMS & TOTAL	CURRENT STATUS	PLACED AT	OPERATOR ACTION
ORD-10042	Aarav Sharma Indiranagar, BLR	3 Items ₹2,499.00 (UPI Paid)	ORDER_PLACED		Confirm Order
ORD-10041	Pooja Hegde	1 Item (₹850.00)	CONFIRMED	Today, 09:45 AM	Start Processing
ORD-10040	Vikram Malhotra	4 Items (₹4,200.00)	PROCESSING	Today, 09:12 AM	Pack Order

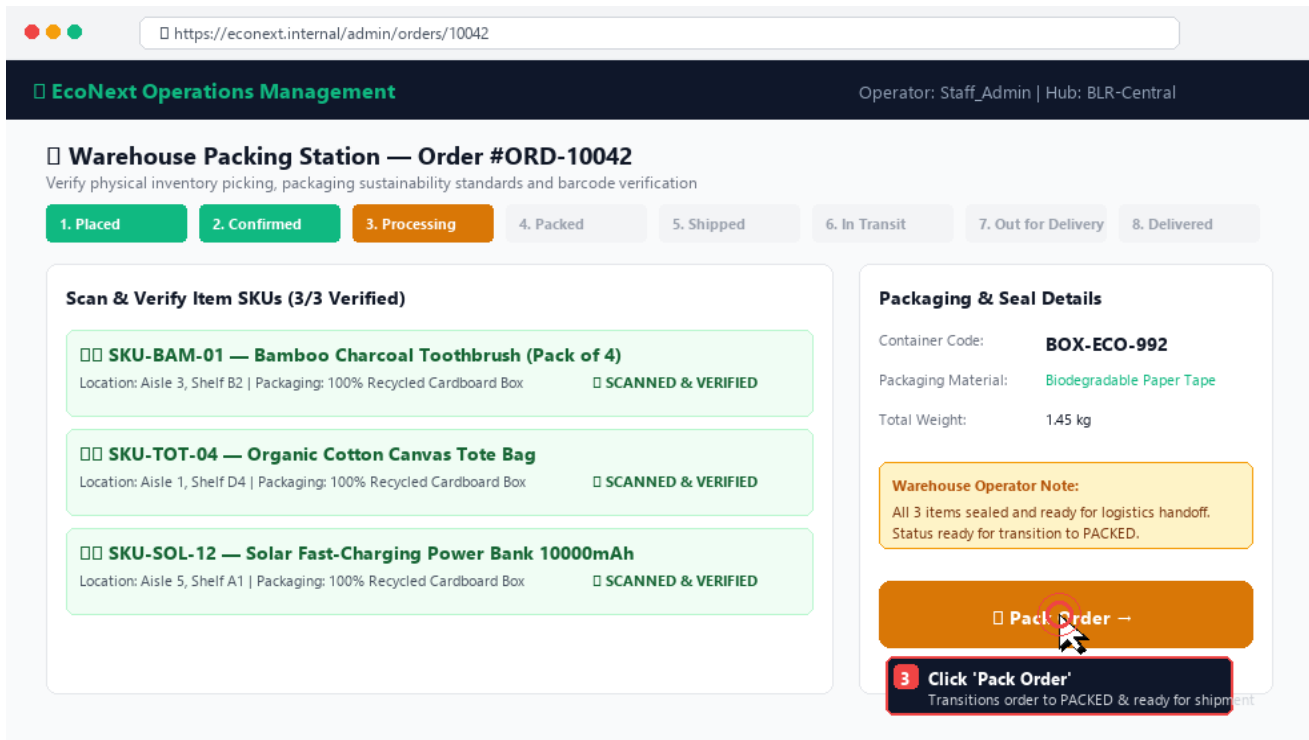
Screenshot 2: Admin Orders Management Portal — Mouse Pointer clicking blue 'Confirm Order' button

■ STATE MACHINE AUTOMATION

When 'Confirm Order' is clicked, the system records the logged-in staff username (e.g., Staff_Admin) in the audit log and fires a Kafka message on order.status.updated. The next allowed action button automatically updates to purple 'Start Processing'.

Step 3: Warehouse Picking, Packing & Packaging Verification

Workflow Description: Warehouse packing personnel open the order detail page. The operator transitions the order to PROCESSING to generate the physical pick-list. After scanning all product SKUs (e.g., Bamboo Toothbrush, Canvas Bag, Solar Charger) and sealing them in an eco-friendly cardboard box (BOX-ECO-992), the operator clicks the orange **'Pack Order'** button to transition status to PACKED.



Screenshot 3: Warehouse Packing Station — Mouse Pointer clicking orange 'Pack Order' button

Warehouse Packaging Standards:

- Use 100% post-consumer recycled corrugated boxes with water-activated paper tape.
- Affix barcode shipping label containing order reference and destination hub code.
- Verify total packed package weight (e.g., 1.45 kg) matches estimated manifest weight within $\pm 5\%$.

Step 4: Fulfillment Hub — Physical Shipment Provisioning

Workflow Description: Once an order reaches PACKED, it appears in the 'Orders Awaiting Fulfillment' queue on the Fulfillment page (/admin/fulfillment). The fulfillment manager opens the 'Provision Physical Shipment' modal dialog, selects the carrier (e.g., EcoExpress Carbon-Neutral), assigns the vehicle number (KA-01-EQ-9124), designates the origin/destination hubs, and clicks **'Create Physical Shipment'**.

EcoNext Operations Management Operator: Staff_Admin | Hub: BLR-Central

Provision Physical Shipment for Order #ORD-10042

Target Order Reference: ORD-10042 (Aarav Sharma - 3 Items)	Logistics Carrier Partner: EcoExpress Carbon-Neutral (Default) ▼
Assigned Vehicle / Fleet No: KA-01-EQ-9124 (EV Delivery Van)	Origin Dispatch Hub: Bengaluru Central Fulfillment Hub
Destination Delivery Hub: Indiranagar Last-Mile Hub, BLR	Estimated Transit Days: 2 - 3 Days (Surface Express)

System Action:
Creating shipment assigns tracking number '#SHP-88210', sets status CREATED, and publishes Kafka event 'shipment.status.updated'.

4 Click 'Create Shipment'
Generates #SHP-88210 in MySQL & emits Kafka event 'shipment.status.updated'.

Screenshot 4: Fulfillment Hub Provisioning Modal — Mouse Pointer clicking 'Create Shipment' button

■ PHYSICAL SHIPMENT PROVISIONING API

Creating a physical shipment calls `POST /api/order-ops/shipments`, which assigns a tracking number (#SHP-88210), creates a record in `operational_shipment` table with status `CREATED`, and links all order items.

Step 5: Shipment Dispatch & Real-Time GPS Telemetry Updates

Workflow Description: As the carrier vehicle departs the fulfillment hub, the shipment transitions from DISPATCHED to IN_TRANSIT. During highway transit or linehaul stops, logistics operators or automated GPS transponders send live milestone coordinates (latitude, longitude, checkpoint name) via the **GPS Telemetry Modal** or the API (PATCH /api/order-ops/fulfillment/shipments/{id}/location).

The screenshot shows the 'Update GPS Coordinates' modal for Shipment #SHP-88210. The modal is titled 'Update GPS Coordinates — Shipment #SHP-88210'. It contains the following information:

- Carrier:** EcoExpress (EV-Van KA-01-EQ-9124)
- Order Ref:** ORD-10042 | **Current Status:** IN_TRANSIT | **Live WebSocket:** CONNECTED
- Latitude Coordinates (e.g. 12.9716):** 12.971598
- Longitude Coordinates (e.g. 77.5946):** 77.594562
- Milestone Location Name:** Hosur Road Expressway Checkpoint, Bengaluru Hub Corridor
- Operator Telemetry Note:** Vehicle passing electronic toll gate. Traffic smooth. ETA 45 mins.
- Broadcasts to:** Kafka 'shipment.location.updated' & STOMP /topic/shipments/88210
- Action:** A red box highlights a button labeled 'Click "Push GPS Telemetry"' with a tooltip that says 'Dispatches live coordinates to customer map'.

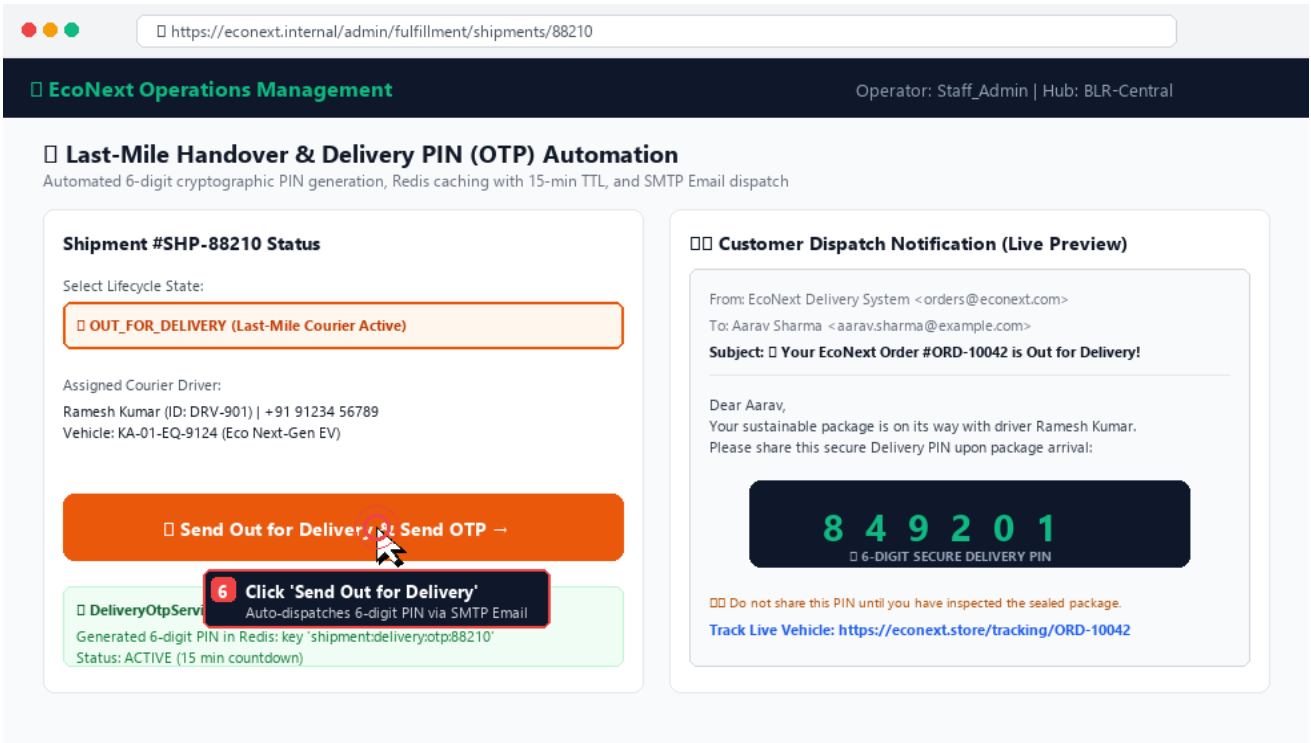
Screenshot 5: Real-Time GPS Telemetry Modal — Mouse Pointer clicking blue 'Push GPS Telemetry' button

GPS Telemetry Data Flow:

1. Operator enters latitude: 12.971598 and longitude: 77.594562 with checkpoint note.
2. Spring Boot persists record into logistics_tracking_event table.
3. Event published to Kafka topic shipment.location.updated.
4. WebSocket STOMP broadcaster pushes payload to subscribed clients at /topic/shipments/{id} and /topic/orders/{orderId} with zero delay (<50ms).

Step 6: 'Out for Delivery' Transition & Automated Delivery OTP Email

Workflow Description: When the vehicle reaches the destination last-mile hub, the shipment is handed to the local courier driver (e.g., Ramesh Kumar, DRV-901). The operator transitions the shipment status to OUT_FOR_DELIVERY. This triggers the **DeliveryOtpService**, which automatically generates a secure 6-digit PIN in Redis and dispatches an HTML email notification to the customer.



Screenshot 6: Out for Delivery & Delivery OTP Notification — Mouse Pointer clicking 'Send Out for Delivery'

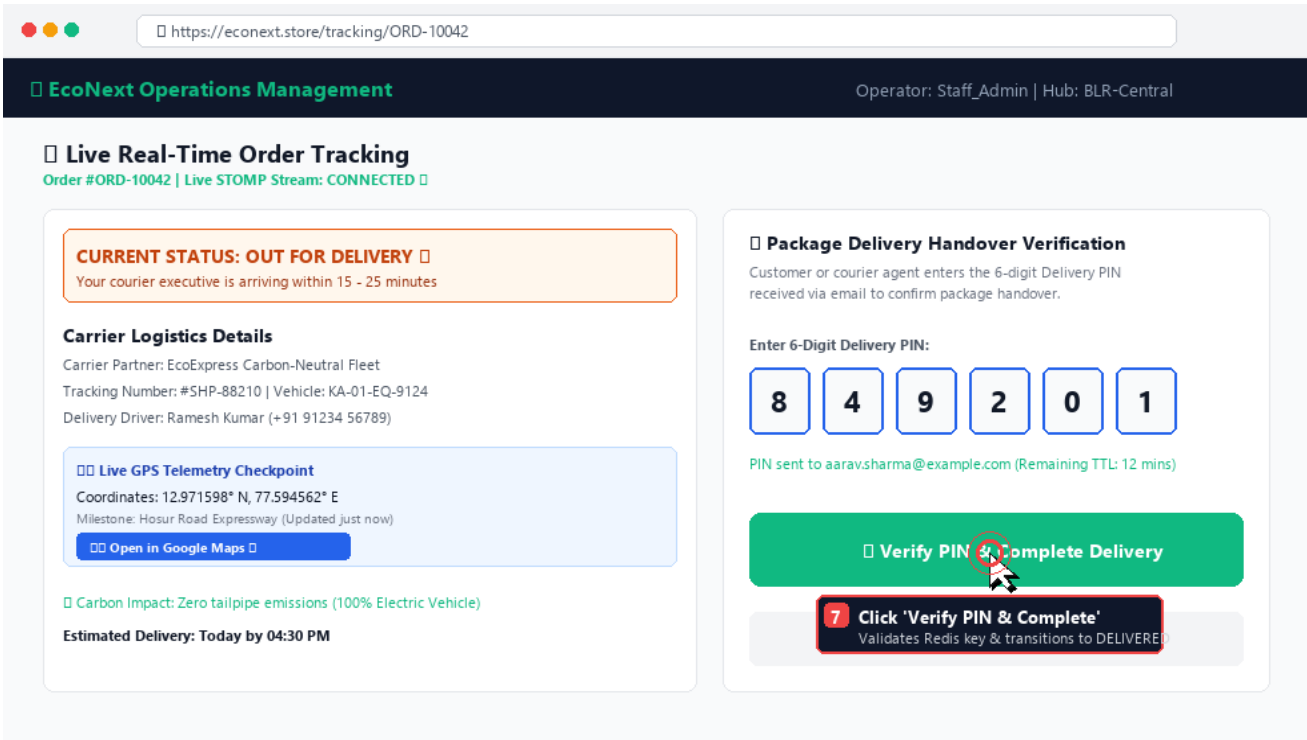
■ SECURITY SPECIFICATION: DELIVERY OTP / PIN

Delivery OTP Specifications:

- Storage: Redis key shipment:delivery:otp:{shipmentId}
- Value: 6-digit cryptographic numeric PIN (e.g., 849201)
- Expiration: 15 minutes (900 seconds) Time-To-Live (TTL)
- Rate Limiting: Maximum 5 verification attempts before lockout

Step 7: Customer Live Tracking Portal & Google Maps Navigation

Workflow Description: The customer accesses their live tracking dashboard at <https://econext.store/tracking/ORD-10042>. The React customer frontend establishes a live STOMP connection to `/ws/tracking`, displaying the real-time vehicle location, carrier info, and a direct clickable button '■■■ Open in Google Maps' (<https://www.google.com/maps?q=12.971598,77.594562>).



Screenshot 7: Customer Live Tracking Screen — Mouse Pointer entering Delivery PIN and clicking 'Verify PIN & Complete Delivery'

Customer Interface Capabilities:

- Real-time progress bar reflecting current status (OUT_FOR_DELIVERY).
- Driver name, phone number, vehicle registration number, and carbon offset rating.
- Direct external link to Google Maps with exact vehicle GPS coordinates.
- Secure 6-digit PIN input box for package handover verification.

Step 8: Secure Handover Verification & Audit Trail Completion

Workflow Description: When the delivery agent arrives at the customer address with the package: 1. The customer shares the 6-digit PIN (849201). 2. The driver or customer enters the PIN and submits verification (POST `/api/order-ops/fulfillment/shipments/{id}/verify-delivery-otp`). 3. Backend verifies the PIN against Redis, transitions shipment to DELIVERED, automatically transitions order to DELIVERED, and writes an immutable audit record.

https://econext.internal/admin/orders/10042/audit

EcoNext Operations Management

Operator: Staff_Admin | Hub: BLR-Central

Order & Shipment Successfully Delivered

Order #ORD-10042 — Full immutable audit timeline with cryptographic OTP verification log

STATUS: DELIVERED (ORDER COMPLETED)

Delivered on Today at 03:42 PM IST | Verified by OTP #849201 | Handed over by Driver Ramesh Kumar (DRV-901)

TIMESTAMP	TRANSITION EVENT	ACTOR / STAFF	DETAILS / AUDIT PROOF	SYSTEM STATUS
15:42:10 IST	OUT_FOR_DELIVERY → DELIVERED	Driver Ramesh (D	8 Audit Timeline Verified Immutable audit trail logged to MySQL database. Delivery OTP generated & emailed.	DELIVERED
14:30:00 IST	IN_TRANSIT → OUT_FOR_DELIVERY	System Automation	Delivery OTP generated & emailed	OUT_FOR_DELIVERY
12:15:30 IST	DISPATCHED → IN_TRANSIT	Hub Operator (OP-44)	EV Van KA-01-EQ-9124 linehaul	IN_TRANSIT
11:00:15 IST	PACKED → DISPATCHED	Warehouse Staff (WH-12)	Assigned to #SHP-88210	DISPATCHED
10:30:00 IST	PROCESSING → PACKED	Warehouse Staff (WH-12)	Scanned 3 SKUs into BOX-ECO-992	PACKED
10:18:45 IST	ORDER_CONFIRMED → PROCESSING	Admin (admin_ops)	Payment verified via Razorpay	PROCESSING
10:15:00 IST	ORDER_PLACED → CONFIRMED	Admin (admin_ops)	Order placed by Aarav Sharma	CONFIRMED

Screenshot 8: Order Completed & Audit Log — Mouse Pointer displaying completed audit trail and verified delivery token

5. Exception Handling, Retries & Failed Deliveries

During physical logistics operations, anomalies such as customer unavailability, incorrect delivery pins, or vehicle breakdowns may occur. Below are standard operating procedures (SOP) for handling edge cases:

Scenario / Exception	Root Cause / Symptom	Standard Operating Procedure (SOP)	System State Transition
Customer Unavailable at Delivery	Customer unreachable by phone or not present at address during delivery attempt.	Driver clicks 'Report Delivery Exception'. Staff enters reason note. Package returned to local hub for re-attempt next business day.	OUT_FOR_DELIVERY → FAILED_DELIVERY
Expired or Missing Delivery PIN	Customer did not receive email or 15-minute Redis TTL expired.	Operator or driver clicks 'Resend Delivery PIN to Email' in portal or triggers POST /shipments/{id}/delivery-otp/send. New PIN generated.	Refreshes Redis TTL to 15 mins; sends new email.
Invalid PIN / Rate Limiting Lockout	5 consecutive incorrect PIN entries submitted.	Redis key temporarily locked. Hub manager must verify customer government ID / order invoice and perform admin authorization override.	Logged to audit log with Admin staff attribution.
Vehicle Breakdown / Route Diversion	Linehaul EV van encounters mechanical failure or road closure.	Operator submits new GPS coordinates with note 'Route diverted via Highway 44'. Customer tracking UI reflects updated ETA automatically.	Status remains IN_TRANSIT; new location logged.

6. Complete Operational API Reference

All operational endpoints are hosted by the **Order Operations Microservice (Port 8083)** and routed via the **Spring Cloud API Gateway (Port 8080)**.

HTTP Method & Route	Required Role / Scope	Request Payload / Parameters	Description & Response
GET /api/order-ops/orders	ORDER_READ	status, search, page, size	Search and filter customer orders with pagination.
PATCH /api/order-ops/orders/{id}/status	ORDER_STATUS_UPDATE	{"nextStatus": "PROCESSING", "note": "..."}	Transition order status with staff attribution in audit log.
POST /api/order-ops/shipments	ORDER_PROCESS	{"targetOrderId": 10042, "carrierName": "...", "vehicleNumber": "..."}	Provision physical shipment and allocate order items.
PATCH /api/order-ops/shipments/{id}/status	ORDER_STATUS_UPDATE	?status=OUT_FOR_DELIVERY	Transition shipment status; auto-triggers OTP if OUT_FOR_DELIVERY.

PATCH /api/order-ops/shipments/{id}/location	ORDER_STATUS_UPDATE	{"latitude": 12.9716, "longitude": 77.5946, "locationName": "..."}"	Post GPS telemetry milestone; broadcasts over Kafka & STOMP.
POST /api/order-ops/shipments/{id}/delivery-otp/send	Public / Staff	None (Path variable shipment ID)	Generate 6-digit PIN in Redis and send HTML email to customer.
POST /api/order-ops/shipments/{id}/delivery-otp/verify	Public / Staff	{"otp": "849201"} or query ?otp=849201	Verify PIN, mark shipment DELIVERED, synchronize order.
GET /api/order-ops/orders/{id}/timeline	ORDER_READ	Path variable order ID	Retrieve immutable audit trail of all historical transitions.

7. Messaging & Live STOMP WebSocket Channels

EcoNext uses Apache Kafka for asynchronous inter-service messaging and STOMP WebSockets for real-time frontend push notifications.

Channel / Topic	Type & Protocol	Publisher	Subscribers / Consumers	Payload Content
order.status.updated	Kafka Event	Order Operations Microservice	Catalog Service, Notification Service, Django	orderId, fromStatus, toStatus, staffId, timestamp
shipment.status.updated	Kafka Event	Order Operations Microservice	WebSocket Gateway, Analytics Engine	shipmentId, orderId, status, carrier, trackingNo
shipment.location.updated	Kafka Event	Order Operations Microservice	WebSocket Gateway Broadcaster	shipmentId, lat, lon, locationName, timestamp
/topic/orders/{orderId}	STOMP WebSocket	WebSocket Gateway	Customer Tracking UI (OrderTrackingPage.js)	Real-time order milestone updates and live GPS pushes.
/topic/shipments/{shipmentId}	STOMP WebSocket	WebSocket Gateway	Admin Fulfillment UI (FulfillmentPage.jsx)	Live vehicle telemetry, hub arrival, and driver events.

8. Operations Staff FAQ & Quick Reference Card

Q: What happens if an operator tries to skip a state (e.g. ORDER_PLACED directly to DELIVERED)?

A: The backend state machine in `OperationalOrderService.java` strictly validates `ALLOWED_NEXT_STATUSES`. Any illegal jump throws an `IllegalStateException` with a 400 Bad Request error. The UI only enables valid next-state buttons.

Q: Can multiple physical shipments be created for a single order?

A: Yes. The `OperationalShipment` entity has a Many-to-One relationship with `OperationalOrder`. If items must ship from different regional warehouses, separate shipments can be provisioned with distinct carriers, tracking numbers, and vehicles.

Q: What happens if the customer does not have access to their email during delivery?

A: The delivery driver or hub operator can click 'Resend Delivery PIN' in the portal. If the customer is completely without internet/email, an authorized Admin operator can inspect the active OTP status endpoint or verify customer identity documents and enter the PIN on behalf of the customer with full audit attribution.

Q: How does the Google Maps link generate in the customer tracking page?

A: When GPS telemetry is updated (latitude, longitude), the customer tracking page (`OrderTrackingPage.js`) dynamically formats the direct Google Maps URL: `https://www.google.com/maps?q={latitude},{longitude}`, allowing one-click navigation.

■ ESCALATION MATRIX & SUPPORT CONTACTS

Operations Support Contacts:

- Central Fulfillment Desk: ext. 4012 | email: ops-hub@econext.internal
- IT & Microservices On-Call: ext. 9901 | Slack: [#econext-prod-ops](#)
- Logistics Fleet Dispatch: ext. 3302 | Emergency Driver Hotline: +91 (800) 555-ECO-1